

Find the error

Instructions for use

- These slides are best viewed in **full screen mode** after downloading the PDF file to your computer.
- Ideally, use a PDF browser without “smooth” transitions.
- Try each exercise yourself first. Then advance forward and **find the error(s)** in the incorrect solution. The next slide has an explanation. The last slide shows an exemplary solution.

Simplify $(x^4)^2 + x^3x^5$

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$$(x^4)^2 + x^3x^5 \rightarrow x^8 + x^8 \rightarrow 2x^8$$

Find the error

Simplify $(x^4)^2 + x^3x^5$

$$(x^4)^2 + x^3x^5 \rightarrow x^8 + x^8 \rightarrow 2x^8$$

Do not use arrows here!

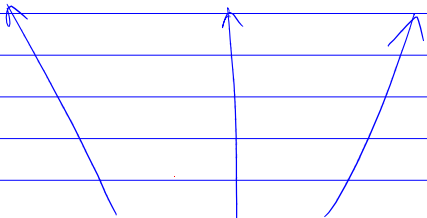
Explanation

Simplify $(x^4)^2 + x^3x^5$

$$(x^4)^2 + x^3x^5 = x^8 + x^8 = 2x^8$$

Correct solution

Simplify $(x^4)^2 + x^3x^5$

$$\boxed{(x^4)^2 + x^3x^5} = \boxed{x^8 + x^8} = \boxed{2x^8}$$


Expressions written between the $=$ signs must be equal.

Additional comment

If $f(x) = 3x^2$, find $f(10)$

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$$f(x) = 3 \cdot 10^2 = 300$$

Find the error

If $f(x) = 3x^2$, find $f(10)$

$$f(x) = 3 \cdot 10^2 = 300$$



In $f(x) = 3x^2$, replace both x 's with 10's.



Explanation

If $f(x) = 3x^2$, find $f(10)$

$$f(10) = 3 \cdot 10^2 = 300$$

Correct solution

Find $\lim_{x \rightarrow 2} (x^3 - 7)$

Find $\lim_{x \rightarrow 2} (x^3 - 7)$

$$\lim_{x \rightarrow 2} (x^3 - 7) \rightarrow 2^3 - 7 \rightarrow 8 - 7 \rightarrow 1$$

Find the error

Find $\lim_{x \rightarrow 2} (x^3 - 7)$

$$\lim_{x \rightarrow 2} (x^3 - 7) \rightarrow 2^3 - 7 \rightarrow 8 - 7 \rightarrow 1$$

$\uparrow$ use = $\uparrow$ use = $\uparrow$ use =

An equal sign (=) should be used to indicate numbers which are equal.

Explanation

Find $\lim_{x \rightarrow 2} (x^3 - 7)$

$$\lim_{x \rightarrow 2} (x^3 - 7) = 2^3 - 7 = 8 - 7 = 1$$

Correct solution

Find $\lim_{x \rightarrow 3} (4x^2 + 2x + 1)$

Find $\lim_{x \rightarrow 3} (4x^2 + 2x + 1)$

$$\lim_{x \rightarrow 3} (4x^2 + 2x + 1) = \lim_{x \rightarrow 3} (4 \cdot 3^2 + 2 \cdot 3 + 1) = 36 + 6 + 1 = 43$$

Find the error

Find $\lim_{x \rightarrow 3} (4x^2 + 2x + 1)$

$$\lim_{x \rightarrow 3} (4x^2 + 2x + 1) = \lim_{x \rightarrow 3} (4 \cdot 3^2 + 2 \cdot 3 + 1) = 36 + 6 + 1 = 43$$

The limit of
the function should
be a number,
not a limit problem

Explanation

Find $\lim_{x \rightarrow 3} (4x^2 + 2x + 1)$

$$\lim_{x \rightarrow 3} (4x^2 + 2x + 1) = 4 \cdot 3^2 + 2 \cdot 3 + 1 = 36 + 6 + 1 = 43$$

$f(x) = 4x^2 + 2x + 1$ is a polynomial.

For a polynomial, $\lim_{x \rightarrow 3} f(x) = f(3)$

Correct solution

If $f(x) = x^2 + 1$, find $\lim_{x \rightarrow 3} f(x)$.

If $f(x) = x^2 + 1$, find $\lim_{x \rightarrow 3} f(x)$.

$$\lim_{x \rightarrow 3} f(x) = x^2 + 1 = 3^2 + 1 = 10$$

Find the error

If $f(x) = x^2 + 1$, find $\lim_{x \rightarrow 3} f(x)$.

$$\lim_{x \rightarrow 3} f(x) = x^2 + 1 = 3^2 + 1 = 10$$

If replacing this with this,
then this is still missing

Additionally, $x^2 + 1$ is not
equal to $3^2 + 1$

Explanation

If $f(x) = x^2 + 1$, find $\lim_{x \rightarrow 3} f(x)$.

$$\boxed{\lim_{x \rightarrow 3} f(x)} = \boxed{x^2 + 1} = \boxed{3^2 + 1} = \boxed{10}$$

Explanation

Be careful with use of equal signs.

Boxes have been placed to identify the content between equal signs.

To use equal signs in this way, each box must represent the same quantity. Here, they do not.

Further explanation of using the equal sign

If $f(x) = x^2 + 1$, find $\lim_{x \rightarrow 3} f(x)$.

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} (x^2 + 1) = 3^2 + 1 = 10$$

Correct solution

Find $\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x}$

Find $\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x}$

$$\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x} = \frac{9+6x+x^2-9}{x} = \frac{6x+x^2}{x} = \frac{x(6+x)}{x} = 6+x = 6$$

Find the error

Find $\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x}$

$$\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x} = \frac{9+6x+x^2-9}{x} = \frac{6x+x^2}{x} = \frac{x(6+x)}{x} = \underbrace{6+x} = \underbrace{6}$$

$6+x$ is not equal to 6

The limit of a function is a number. This, however, is not a number, but is rather a function!

(The limit of a function is not a function!)

Explanation

Find $\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x}$

$$\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x} = \lim_{x \rightarrow 0} \frac{9 + 6x + x^2 - 9}{x} = \lim_{x \rightarrow 0} \frac{6x + x^2}{x} = \lim_{x \rightarrow 0} \frac{x(6+x)}{x} = \lim_{x \rightarrow 0} (6+x) = 6$$

Correct solution

Find $\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x}$

$$\lim_{x \rightarrow 0} \frac{(3+x)^2 - 9}{x} = \lim_{x \rightarrow 0} \frac{9 + 6x + x^2 - 9}{x} = \lim_{x \rightarrow 0} \frac{6x + x^2}{x} = \lim_{x \rightarrow 0} \frac{x(6+x)}{x} = \lim_{x \rightarrow 0} (6+x) = 6$$

The limit of this function is a number

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The limit of this function is a number

This is
the number.

Each equals sign indicates that
two numbers are equal.

The numbers are circled
blue.

Explanation of correct solution

Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$

Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5} = \frac{5^2 - 25}{5 - 5} = \frac{0}{0} \leftarrow \text{undefined}$$

Find the error

Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5} = \frac{5^2 - 25}{5 - 5} = \frac{0}{0} \leftarrow \text{undefined}$$

$\lim_{x \rightarrow 5} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow 5} f(x)}{\lim_{x \rightarrow 5} g(x)}$ can only be used if $\lim_{x \rightarrow 5} g(x)$ is not zero $\leftarrow$
(and if both $\lim_{x \rightarrow 5} f(x)$ and $\lim_{x \rightarrow 5} g(x)$ are numbers).

Unfortunately, $\lim_{x \rightarrow 5} g(x)$ is zero

Explanation

Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5} = \lim_{x \rightarrow 5} \frac{(x+5)\cancel{(x-5)}}{\cancel{x-5}} = \lim_{x \rightarrow 5} (x+5) = 5+5 = 10$$

(x+5 is polynomial)

If $\lim_{x \rightarrow 5^-} f(x)$ is zero and $\lim_{x \rightarrow 5^+} g(x)$ is zero
then use algebra to cancel a common factor.

Correct solution

Find $\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x - 2}$

Find $\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x - 2}$

$$\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x - 2} = \frac{2^2 + 3 \cdot 2 - 10}{2 - 2} = \frac{0}{0}$$

Find the error

Find $\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x - 2}$

$$\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x - 2} = \frac{2^2 + 3 \cdot 2 - 10}{2 - 2} = \frac{0}{0}$$

→ Incorrect!

The 'skipped step' here is:

$$\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x - 2} = \frac{\lim_{x \rightarrow 2} (x^2 + 3x - 10)}{\lim_{x \rightarrow 2} (x - 2)} = \frac{2^2 + 3 \cdot 2 - 10}{2 - 0}$$

Annotations: "these are equal" (pointing to the first and second fractions), "these are equal" (pointing to the second and third fractions).

You cannot just always plug in for x.

$\lim_{x \rightarrow 2} f(x)$ is not always the same as $f(2)$

THIS is invalid!

To use $\lim_{x \rightarrow 2} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow 2} f(x)}{\lim_{x \rightarrow 2} g(x)}$, you must ensure $\lim_{x \rightarrow 2} g(x)$ is not zero.

Here, $\lim_{x \rightarrow 2}$ is zero!!

Find $\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x - 2}$

$$\lim_{x \rightarrow 2} \frac{x^2 + 3x - 10}{x - 2} = \lim_{x \rightarrow 2} \frac{(x+5)(x-2)}{x-2} = \lim_{x \rightarrow 2} (x+5) = 2+5 = 7$$

Correct solution

Find $\lim_{x \rightarrow -2} \frac{x^2 + 7x + 10}{x + 2}$

Find $\lim_{x \rightarrow -2} \frac{x^2 + 7x + 10}{x + 2}$

$$\lim_{x \rightarrow -2} \frac{x^2 + 7x + 10}{x + 2} \left(\frac{0}{0} \right) = \lim_{x \rightarrow -2} \frac{(x+5)(x+2)}{x+2} = \lim_{x \rightarrow -2} (x+5) = -2 + 5 = 3$$

Find the error

Find $\lim_{x \rightarrow -2} \frac{x^2 + 7x + 10}{x + 2}$

$$\lim_{x \rightarrow -2} \frac{x^2 + 7x + 10}{x + 2} \left(\frac{0}{0} \right) = \lim_{x \rightarrow -2} \frac{(x+5)(x+2)}{x+2} = \lim_{x \rightarrow -2} (x+5) = -2 + 5 = 3$$

What is this doing here?

Writing $\left(\frac{0}{0} \right)$ here makes it look like you're multiplying this by $\frac{x^2 + 7x + 10}{x + 2}$.

This breaks the "limit of a function is a number" notation/idea

Find $\lim_{x \rightarrow -2} \frac{x^2 + 7x + 10}{x + 2}$

$$\lim_{x \rightarrow -2} \frac{x^2 + 7x + 10}{x + 2} = \lim_{x \rightarrow -2} \frac{(x+5)(x+2)}{x+2} = \lim_{x \rightarrow -2} (x+5) = -2 + 5 = 3$$

Correct solution

Let $f(x) = \frac{x}{|x|}$ and find $\lim_{x \rightarrow 0} f(x)$

Let $f(x) = \frac{x}{|x|}$ and find $\lim_{x \rightarrow 0} f(x)$

$$\lim_{x \rightarrow 0^+} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = -1$$

$\lim_{x \rightarrow 0} f(x)$ does not exist because $\lim_{x \rightarrow 0^+} \neq \lim_{x \rightarrow 0^-}$

Find the error

Let $f(x) = \frac{x}{|x|}$ and find $\lim_{x \rightarrow 0} f(x)$

$$\lim_{x \rightarrow 0^+} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = -1$$

$\lim_{x \rightarrow 0} f(x)$ does not exist because $\lim_{x \rightarrow 0^+} \neq \lim_{x \rightarrow 0^-}$

↑ ↑
The limit of a function is a number.
Where's the function?

(Write either $\boxed{f(x)}$ or $\boxed{\frac{x}{|x|}}$)

Explanation

Let $f(x) = \frac{x}{|x|}$ and find $\lim_{x \rightarrow 0} f(x)$

$$\lim_{x \rightarrow 0^+} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = -1$$

$\lim_{x \rightarrow 0} f(x)$ does not exist because $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$

Correct solution

The end
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